
Part 4: Game Theory Practice Problems

Econ 502: Advanced Microeconomics

Normal Form Games

1. Dominant Strategies and Nash Equilibrium

Consider the following 3×3 simultaneous-move game. Player A is the row player, Player B is the column player.

	L	C	R
U	(4, 3)	(5, 1)	(6, 2)
M	(2, 1)	(8, 4)	(3, 6)
D	(3, 0)	(9, 6)	(2, 8)

- Recall that a strategy is **strictly dominated** if some other strategy yields a strictly higher payoff regardless of what the opponent does. Identify any strictly dominated strategy in this game and the strategy that dominates it.
- Using best-response analysis, find all pure-strategy Nash equilibria of the original game.
- Suppose each player believes their opponent is rational and so will not play a strictly dominated strategy. Use the dominance you found in (a) to eliminate one strategy, then re-examine the resulting smaller game. If a further strategy becomes strictly dominated, eliminate that too and continue. What outcome does this reasoning predict, and how does it relate to your answer in (b)?
- Strict dominance is a *stronger* requirement than Nash equilibrium: a strictly dominant-strategy equilibrium is always a Nash equilibrium, but the converse fails. Give an example of a Nash-equilibrium strategy in the game above that is **not** strictly dominant for the player who plays it, and explain why this does not contradict its being part of a Nash equilibrium.

2. Battle of the Sexes (Mixed Strategies)

A couple is deciding where to spend the evening. Player A prefers the **Opera**, Player B prefers the **Football** game, but both would rather be together than alone.

	B: Opera	B: Football
A: Opera	(3, 1)	(0, 0)
A: Football	(0, 0)	(1, 3)

- Find the two pure-strategy Nash equilibria. Which player prefers each equilibrium? Why is this game called a “coordination game with conflict”?
- Find the mixed-strategy Nash equilibrium. Let p be the probability A plays Opera and q be the probability B plays Opera. Derive p^* and q^* .
- Compute each player’s expected payoff in the mixed-strategy equilibrium. How does it compare to either of the pure-strategy equilibria? Why might players prefer to coordinate on a pure equilibrium even if the choice favors the other player?
- Suppose A can commit (publicly and credibly) to going to the Opera before B chooses. What is the equilibrium outcome? What does this illustrate about the value of commitment?

3. Stag Hunt: Risk vs. Payoff Dominance

Two hunters can either cooperate to hunt a stag (large payoff but requires both) or hunt a hare alone (small payoff, guaranteed).

	B: Stag	B: Hare
A: Stag	(5, 5)	(0, 4)
A: Hare	(4, 0)	(4, 4)

- Find all pure-strategy Nash equilibria. Show that one equilibrium is **payoff-dominant** (both players prefer it) and another is **risk-dominant** (each player’s best response if they think the opponent is equally likely to choose either action).
- Find the mixed-strategy Nash equilibrium. What is the probability each player puts on Stag?
- Suppose A believes B will choose Stag with probability π . For what values of π does A play Stag? Use this to discuss why coordination failure (both choosing Hare) can be a stable outcome even though both players prefer (Stag, Stag).
- The Stag Hunt is sometimes used as a metaphor for collective action problems (e.g., environmental treaties, vaccinations, financial crises). Pick one example and explain how the Stag Hunt structure applies.

Sequential Games and Commitment

4. Entry Deterrence with Capacity Commitment

An incumbent firm (I) faces a potential entrant (E). The game has two stages:

- The incumbent chooses **capacity** $k \in \{ \text{Low}, \text{High} \}$ before E moves.
- E observes k and chooses to **Enter** or **Stay Out**. If E enters, both firms compete in the market.

The payoffs (Entrant, Incumbent) for each combination are:

Capacity	E: Stay Out	E: Enter
Low	(0, 12)	(3, 6)
High	(0, 8)	(-2, 4)

(Building capacity is a sunk cost for the incumbent — High capacity always yields lower profit for I than Low. But High capacity also commits the incumbent to flood the market post-entry, hurting the entrant enough that E stays out.)

- Draw the game tree (extensive form).
- Solve the game by backward induction. What is the subgame perfect equilibrium? What is the equilibrium outcome (capacity choice and entry decision)?
- Suppose the incumbent could not commit to High capacity (i.e., capacity could be changed costlessly after E's decision). What would happen? Why does the ability to commit to High capacity benefit the incumbent?
- Connect this game to the concept of a **non-credible threat**. If the incumbent simply *threatened* aggressive post-entry competition (without building capacity), would E believe it? Explain.

5. Ultimatum Bargaining

Two players bargain over \$10. Player 1 (the proposer) offers a split (s_1, s_2) where $s_1 + s_2 = 10$ and $s_2 \geq 0$. Player 2 (the responder) either accepts (each gets their proposed share) or rejects (both get 0). Assume s_2 can take any nonnegative real value (in dollars or cents).

- Solve the game by backward induction. What does Player 2 do when offered $s_2 > 0$? What does Player 2 do when offered $s_2 = 0$? Assume that when indifferent, Player 2 accepts.
- What is Player 1's optimal offer in the subgame perfect equilibrium? What are the equilibrium payoffs?
- Empirically, in laboratory ultimatum games, responders frequently reject offers below about 30% of the pie, and proposers often offer close to 50–50. How does this evidence challenge the standard prediction? Suggest one modification of the standard model that could rationalize the observed behavior.
- Now suppose the game has **two rounds**: if Player 2 rejects in round 1, Player 2 makes a counteroffer in round 2 over a pie that has shrunk to $\delta \cdot 10$ (where $\delta \in (0, 1)$ is a discount factor reflecting the cost of delay). If Player 1 rejects the round-2 offer, both get 0. Solve by backward induction. How much does Player 1 offer in round 1? How does the equilibrium depend on δ ?

Repeated Games

6. Repeated Prisoner's Dilemma and Grim Trigger

Two firms play the following stage game in each period:

	B: Cooperate	B: Defect
A: Cooperate	(4, 4)	(0, 5)
A: Defect	(5, 0)	(1, 1)

- Confirm that (Defect, Defect) is the unique Nash equilibrium of the one-shot game.
- Suppose the game is repeated infinitely with discount factor $\delta \in (0, 1)$. Each player considers the **grim trigger** strategy: “Cooperate in period 1, and continue to cooperate as long as the opponent has always cooperated; defect forever after the first deviation.”

Compute the present discounted value of (i) always cooperating and (ii) deviating in period 1 (defect once and trigger permanent defection thereafter). Find the **critical discount factor** δ^* above which grim trigger sustains cooperation as a subgame perfect equilibrium.
- Finitely repeated game.** Now suppose the game is repeated $T < \infty$ times. Show that defection in every period is the unique subgame perfect equilibrium. (Hint: argue by backward induction starting from period T .)
- Why does the prediction differ between the finitely and infinitely repeated games? What real-world feature do infinitely repeated games capture that finite ones do not?

7. Trigger Strategies in a Cournot Duopoly

Two firms produce a homogeneous good with inverse demand $P = 100 - Q$ where $Q = q_1 + q_2$. Both have constant marginal cost $c = 20$. Each period they simultaneously choose quantities, and the stage game is the standard Cournot game.

- Find (i) the Cournot Nash quantities, price, and per-firm profit and (ii) the joint-monopoly (collusive) quantities, price, and per-firm profit (assuming firms split monopoly output equally).
- Suppose firms play the infinitely repeated game with discount factor δ and use the following grim-trigger strategy: produce the collusive quantity each period as long as both firms have always done so; revert to the Cournot Nash quantity forever upon any deviation.

Compute the **best-deviation profit**: the one-period payoff if firm 1 deviates optimally, taking firm 2's collusive output as given. Then derive the critical δ^* above which collusion is sustainable.
- Suppose firms suspect each other of cheating (so δ effectively falls). What does the model predict about the sustainability of collusion when firms become impatient or when the future is uncertain?
- Real cartels (e.g., OPEC, vitamin cartels) often break down. Identify two features of real markets that make collusion harder to sustain than the simple model predicts.